UNIT 2

THE TOP MYTHS ABOUT ADVANCED AI

A captivating conversation is taking place about the future of artificial intelligence and what it will/should mean for humanity. There are fascinating controversies where the world's leading experts disagree, such as: AI's future impact on the job market; if/when human-level AI will be developed; whether this will lead to an intelligence explosion; and whether this is something we should welcome or fear. But there are also many examples of boring pseudo-controversies caused by people misunderstanding and talking past each other. To help ourselves focus on the interesting controversies and open questions — and not on the misunderstandings — let's clear up some of the most common myths.

Timeline myths

The first myth regards the timeline: how long will it take until machines greatly supersede human-level intelligence? A common misconception is that we know the answer with great certainty.

One popular myth is that we know we'll get superhuman AI this century. In fact, history is full of technological over-hyping. Where are those fusion power plants and flying cars we were promised we'd have by now? AI has also been repeatedly over-hyped in the past, even by some of the founders of the field. For example, John McCarthy (who coined the term "artificial intelligence"), Marvin Minsky, Nathaniel Rochester and Claude Shannon wrote this overly optimistic forecast about what could be accomplished during two months with stone-age computers: "We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College [...] An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer."

On the other hand, a popular counter-myth is that we know we won't get superhuman AI in this century. Researchers have made a wide range of estimates for how far we are from superhuman AI, but we certainly can't say with great confidence that the probability is zero this century, given the dismal track record of such techno-skeptic predictions. For example, Ernest Rutherford, arguably the greatest nuclear physicist of his time, said in 1933 — less than 24 hours before Szilard's invention of the nuclear chain reaction — that nuclear energy was "moonshine." And Astronomer Royal Richard Woolley called interplanetary travel "utter bilge" in 1956. The most extreme form of this myth is that superhuman AI will never arrive because it's physically impossible. However, physicists know that a brain consists of quarks and electrons arranged to act as a powerful computer, and that there's no law of physics preventing us from building even more intelligent quark blobs.

There have been a number of surveys asking AI researchers how many years from now they think we'll have human-level AI with at least 50% probability. All these surveys have the same conclusion: the world's leading experts disagree, so we simply don't know.

There's also a related myth that people who worry about AI think it's only a few years away. In fact, most people on record worrying about superhuman AI guess it's still at least decades away. But they argue that as long as we're not 100% sure that it won't happen this century, it's smart to start safety research now to prepare for the eventuality. Many of the safety problems associated with human-level AI are so hard that they may take decades to solve. So it's prudent to start researching them now rather than the night before some programmers drinking Red Bull decide to switch one on.

Controversy myths

Another common misconception is that the only people harboring concerns about AI and advocating AI safety research are luddites who don't know much about AI. When Stuart Russell, author of the standard AI textbook, mentioned this during his Puerto Rico talk, the audience laughed loudly. A related misconception is that supporting AI safety research is hugely controversial. In fact, to support a modest investment in AI safety research, people don't need to be convinced that risks are high, merely non-negligible — just as a modest investment in home insurance is justified by a non-negligible probability of the home burning down.

It may be that the media have made the AI safety debate seem more controversial than it really is. After all, fear sells, and articles using out-of-context quotes to proclaim imminent doom can generate more clicks than nuanced and balanced ones. As a result, two people who only know about each other's positions from media quotes are likely to think they disagree more than they really do.

Myths about the risks of superhuman AI

Many AI researchers roll their eyes when seeing this headline: "Stephen Hawking warns that rise of robots may be disastrous for mankind." And as many have lost count of how many similar articles they've seen. Typically, these articles are accompanied by an evil-looking robot carrying a weapon, and they suggest we should worry about robots rising up and killing us because they've become conscious and/or evil. On a lighter note, such articles are actually rather impressive, because they succinctly summarize the scenario that AI researchers don't worry about. That scenario combines as many as three separate misconceptions: concern about consciousness, evil, and robots.

If you drive down the road, you have a subjective experience of colors, sounds, etc. But does a self-driving car have a subjective experience? Does it feel like anything at all to be a self-driving car? Although this mystery of consciousness is interesting in its own right, it's irrelevant to AI risk. If you get struck by a driverless car, it makes no difference to you whether it subjectively feels conscious. In the same way, what will affect us humans is what superintelligent AI does, not how it subjectively feels.

The fear of machines turning evil is another red herring. The real worry isn't malevolence, but competence. A superintelligent AI is by definition very good at attaining its goals, whatever they may be, so we need to ensure that its goals are aligned with ours. Humans don't generally hate ants, but we're more intelligent than they are – so if we want to build a hydroelectric dam and there's an anthill there, too bad for the ants. The beneficial-AI movement wants to avoid placing humanity in the position of those ants.

The consciousness misconception is related to the myth that machines can't have goals. Machines can obviously have goals in the narrow sense of exhibiting goal-oriented behavior: the behavior of a heat-seeking missile is most economically explained as a goal to hit a target. If you feel threatened by a machine whose goals are misaligned with yours, then it is precisely its goals in this narrow sense that troubles you, not whether the machine is conscious and experiences a sense of purpose. If that heat-seeking missile were chasing you, you probably wouldn't exclaim: "I'm not worried, because machines can't have goals!"

I sympathize with different robotics pioneers who feel unfairly demonized by scaremongering tabloids, because some journalists seem obsessively fixated on robots and adorn many of their articles with evil-looking metal monsters with red shiny eyes. In fact, the main concern of the beneficial-AI movement isn't with robots but with intelligence itself: specifically, intelligence whose goals are misaligned with ours. To cause us trouble, such misaligned superhuman intelligence needs no robotic body, merely an internet connection – this may enable outsmarting financial markets, out-inventing human researchers, out-manipulating human leaders, and developing weapons we cannot even understand. Even if building robots were physically impossible, a super-intelligent and super-wealthy AI could easily pay or manipulate many humans to unwittingly do its bidding.

The robot misconception is related to the myth that machines can't control humans. Intelligence enables control: humans control tigers not because we are stronger, but because we are smarter. This means that if we cede our position as smartest on our planet, it's possible that we might also cede control.

Exercises

1. Answer the following questions.

1. What are the controversies about the future of artificial intelligence mostly based on?
2. What is the myth relating to the AI in terms of the timeline (explain and give examples)?
3. What is a popular counter-myth relating to the AI in terms of the timeline (explain and give examples)?
4. Why is it smart to start the safety research relating to the AI now?
5. What is another common misconception regarding the people harboring concerns about AI and advocating AI safety research?
6. In which way have media made the AI safety debate seem more controversial than it really is?
7. What are the three misconceptions relating to the myths about the risks of superhuman AI?
8. Can AI have a subjective experience?
9. Can AI be evil?
10. What is the AI consciousness misconception related to? Explain.
11. Explain the following statement: "In fact, the main concern of the beneficial-AI movement isn't with robots but with intelligence itself: specifically, intelligence whose goals are misaligned with ours."
12. What myth is the robot misconception related to?

2. Explain/describe the following terms and expressions, on the basis of their use in the text. Use a dictionary or check online if necessary.

misunderstanding	
technological over-hyping	
stone-age computers	
counter-myth	
techno-skeptic predictions	
human-level AI	
non-negligible probability	
out-of-context quotes	
goal-oriented behavior	
scaremongering tabloids	

3. Connect the sentence parts in accordance with the text and its order to get correct sentences.

The first myth regards the timeline: how long will it take	we won't get superhuman AI in this century.
A popular counter-myth is that we know	to the myth that machines can't have goals.
The most extreme form of this myth is that	the myth that machines can't control humans.
There's also a related myth that people	until machines greatly supersede human-level intelligence?
Another common misconception is that the only people harboring concerns about AI	superhuman AI will never arrive because it's physically impossible.
The fear of machines having consciousness	who worry about AI think it's only a few years away.
The consciousness misconception is related	and advocating AI safety research are luddites who don't know much about AI.
The robot misconception is related to	or turning evil is another myth.

4. Complete the text with the given words and expressions.

| machines | technology | algorithms | human workforces | automation | AI-equipped | "human touch"| consistent | scans and x-rays | artificial intelligence | lightning speed | automated |augmenting | misconceptions | industrial revolutions |

5 MYTHS ABOUT ARTIFICIAL INTELLIGENCE (AI) YOU MUST STOP BELIEVING

Very few subjects in science and ________________ are causing as much excitement right now as _______________ (AI). In a lot of cases this is a good reason, as some of the world's brightest minds have said that it's potential to revolutionise all aspects of our lives is unprecedented.

On the other hand, as with anything new, there are certainly snake-oil salesmen looking to make a quick buck on the basis of promises which can't (yet) be truly met. And there are others, often with vested interests, with plenty of motive for spreading fear and distrust.

So here is a run-through of some basic_______________, and frequently peddled mistruths, which often come up when the subject is discussed, as well as reasons why you shouldn't necessarily buy into them.

AI is going to replace all jobs

It's certainly true that the advent of AI and _____________ has the potential to seriously disrupt labour – and in many situations it is already doing just that. However, seeing this as a straightforward transfer of labour from humans to ___________ is a vast over-simplification.

Previous ___________ have certainly led to transformation of the employment landscape, such as the mass shift from agricultural work to factories during the nineteenth century. The number of jobs (adjusted for the rapid growth in population) has generally stayed ___________ though. And despite what doom-mongers have said there's very little actual evidence to suggest that mass unemployment or widespread redundancy of ___________ is likely. In fact, it is just as possible that a more productive economy, brought about by the increased efficiency and reduction of waste that automation promises, will give us more options for spending our time on productive, income-generating pursuits.

In the short-term, employers are generally looking at AI technology as a method of ___________ human workforces, and enabling them to work in newer and smarter ways.

Only low-skilled and manual workers will be replaced by AI and automation

This is certainly a fallacy. Already, ___________ robots and machinery are carrying out work generally reserved for the most highly trained and professional members of society, such as doctors and lawyers. True, a lot of their focus has been on reducing the "drudgery" of day-to-day aspects of the work. For example, in the legal field, AI is used to scan thousands of documents at___________, drawing out the points which may be relevant in an ongoing case. In medicine, machine learning ___________ assess images such as___________, looking for early warning signs of disease, which they are proving highly competent at spotting. Both fields, however, as well as many other professions, involve a combination of routine, though technically complex, procedures – which are likely to be taken up by machines – as well as ___________ procedures. For a lawyer this could be presenting arguments in court in a way that will convince a jury, and in medicine, it could be breaking news in the most considerate and helpful way. These aspects of the job are less likely to be automated, but members of their respective professions could find they have more time for them – and therefore become more competent at them – if mundane drudgery is routinely___________.

| natural intelligences | autonomous | applications | computers | misconception | | super-smart machines | science fiction | control | autonomously | self-preservation | applying | sci-fi scenarios | super-humans | allowances | human creators |

Super-intelligent computers will become better than humans at doing anything we can do

Broadly speaking, AI ___________ are split into two groups – specialised and generalised. Specialised AIs – ones focused on performing one job, or working in one field, and becoming increasingly good at it – are a fact of life today – the legal and medical applications mentioned above are good examples.

Generalised AIs on the other hand – those which are capable of___________ themselves to a number of different tasks, just as human or ___________ are – are somewhat further off. This is why although we may regularly come across AIs which are better than humans at one particular task, it is likely to be a while before we come face-to-face with robots in the mould of Star Trek's Data –essentially ___________ who can beat us at pretty much anything.

Artificial intelligence will quickly overtake and outpace human intelligence

This is a ___________ brought about by picturing intelligence as a linear scale – for example, from one to 10 – imagining that perhaps animals score at the lower end, humans at the higher end, and with ___________ at the top of the scale.

In reality, intelligence is measured in many different dimensions. In some of them (for example speed of calculations or capacity for recall) ___________ already far outpace us, while in others, such as creative ability, emotional intelligence (such as empathy) and strategic thinking, they are still nowhere near and aren't likely to be any time soon.

AI will lead to the destruction of enslavement of the human race by superior robotic beings

This one is obviously out of any number of___________ – The Terminator and The Matrix are probably the most frequently cited! However, some voices which have proven themselves to be worth listening to in the past – such as physicist Stephen Hawking and tech entrepreneur Elon Musk – have made it very clear they believe the danger is real.

The fact is though, that notwithstanding the distant future, where indeed anything is possible, a great number of boundaries would have to be broken down, and ___________ made by society, before we would be in a position where this would be possible. Right now, it's highly unlikely anyone would think about building or deploying an ___________ machine with the potential to "make up its mind" to hurt and turn against its___________. Although drones and security robots designed to detect and prevent threats, and even take autonomous action to neutralise them, have been developed, they have yet to be deployed and doing so is likely to provoke widespread public condemnation. The hypothetical scenario tends to be that robots either develop ___________ instincts, or re-interpret commands to protect or preserve human life to mean that humans should be taken under robotic___________. As it is unlikely that anyone would build machines with the facilities to carry out these actions___________, this is unlikely to be an immediate problem. Could it happen in the future? It's a possibility, but if you're going to worry about ___________ threats, then it's just as likely that invading aliens will get to us first.